ALI ELRIDA SHAHIMI

Master's Student in Computer Engineering | Graph ML, Knowledge Graphs, and Spatiotemporal AI

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PROFILE

Research-oriented Computer Engineering master's student at Politecnico di Torino and currently an exchange M2 student at INP Grenoble. Strong background in Artificial Intelligence, Machine Learning, Deep Learning, and GPU Computing, with growing research experience in Graph Neural Networks, Knowledge Graphs, and spatiotemporal representation learning. Currently conducting a research internship on Graph Neural Networks for SpatioTemporal Knowledge Graphs representing semantic trajectories, with interests in hybrid symbolic-neural AI, self-supervised learning, multimodal learning, and scientific machine learning.

EDUCATION

Master's Degree in Computer Engineering (LM-32) — *2024–2026*
Politecnico di Torino
Relevant coursework: Machine Learning and Pattern Recognition, Big Data Processing, Software Engineering, System and Device Programming, Web Applications, Computer Networks.

Exchange Program, Semester 9–10 — INP Grenoble *2025–2026*
Relevant coursework: Generative Multimodal AI, GPU Computing, Computer Vision, Information Security, Research Project (S10), Basic Machine Learning Models.

Bachelor's Degree in Computer Engineering — *2021-2024*
Politecnico di Torino *Completed*
Final grade: 104 / 110.

RESEARCH EXPERIENCE

Research Internship — Graph Neural Networks for SpatioTemporal Knowledge Graphs — LIG, Stories Team, Grenoble *2026*

- Studying the contribution of Graph Neural Networks to Knowledge Graphs, with a focus on spatiotemporal knowledge graphs representing semantic trajectories.
- Building ontology-guided graph snapshots from RDF / Knowledge Graph data for downstream graph learning tasks.
- Developing and evaluating GNN and spatiotemporal models for self-supervised representation learning, clustering, and entity alignment analysis.
- Exploring hybrid symbolic-neural approaches combining semantic web structures with deep learning models.

TECHNICAL SKILLS

Programming	Python, C, C++, Java, JavaScript, TypeScript
Frameworks	PyTorch, NumPy, CUDA, TensorFlow, OpenCV, NetworkX, rdflib
AI & ML	Neural Networks, Deep Learning, LLMs, NLP, Graph Neural Networks, Representation Learning, Self-Supervised Learning
KG / Semantic Web	RDF, OWL, SPARQL, Knowledge Graphs, Semantic Trajectories
Web & Software	HTML, CSS, SQLite, React, Node.js, Software Architecture, Web Applications
Tools	Git, Linux, Jupyter, VS Code

SELECTED PROJECTS

Project portfolio: github.com/Ali339275

- **Spatio-Temporal-GNN** — Research-oriented project on spatial and temporal Graph Neural Networks for semantic trajectories represented in Knowledge Graphs.
- **word2vec-cuda** — CUDA implementation of Word2Vec with focus on GPU programming and parallel computation.
- **DDPM / diffusion modeling** — Implementation study of denoising diffusion probabilistic models.
- **GeoControl** — Team-based IoT monitoring system for real-time environmental data visualization.
- **Stuff-Happens** — Full-stack web application using React and Node.js.
- **Other GitHub projects** — Computer vision, segmentation, 3D shape modeling, augmented reality, embedded systems, and data-analysis tools.

RESEARCH INTERESTS

Graph Neural Networks and Knowledge Graph Learning • Spatiotemporal Representation Learning • Semantic Web and Hybrid Symbolic-Neural AI • Self-Supervised and Multimodal Learning • Generative AI and Diffusion Models • GPU and Parallel Computing

CERTIFICATION

IBM AI0201EN: Generative AI Models for Natural Language (NLP & NLU) — Verified Certificate, edX / IBM, October 2025.

LANGUAGES

Arabic: Native | English: Fluent | Italian: Basic